



MLU-EEP - Agentic AI Applications Embark Guide

Basic information

Course code name: MLU-EEP-AGAI-M2

Course name: Agentic AI Applications: from Concept to Code

Projected full duration: 4 hours (Can vary depending to modularization based on student's personas)

Note: Use this document alongside for more references about the course

Embark Guide Motivation

As artificial intelligence rapidly transforms how we build software and solve complex problems, **Agentic AI** represents the next frontier—systems that can reason, plan, and act autonomously to achieve goals. This Embark Guide supports the MLU-EEP Agentic AI Applications course (MLU-EEP-AGAI-M2), a 4-hour foundational module designed to equip educators with the essential concepts and hands-on experience needed to teach Agentic AI effectively.

The motivation for this guide stems from three key needs:

Bridging the Knowledge Gap: Agentic AI is an emerging field that many educators are encountering for the first time. This guide provides the scaffolding needed to confidently introduce these concepts to students across diverse technical backgrounds.

Enabling Flexible Delivery: Educational contexts vary widely—from engineering programs requiring deep technical implementation to business programs focusing on strategic applications. This guide's modular structure allows you to adapt content to your specific audience and time constraints.

Accelerating Practical Understanding: Rather than purely theoretical instruction, this guide emphasizes hands-on learning through persona-based activities. Students gain immediate, practical experience with Agentic AI tools, making abstract concepts tangible and relevant to their future careers.

Whether you're teaching computer science students who will build these systems, business students who will deploy them strategically, or interdisciplinary learners who need foundational literacy, this guide provides the framework to deliver impactful Agentic AI education.

Note: Read the module's Educator Guide before going through this embark guide

How to use this guide

This Embark Guide is designed as a **flexible teaching companion** that works alongside the core course materials. Follow these steps to maximize its effectiveness:

Before You Begin

Read the Module Design Doc first - This provides essential context about learning objectives, assessment strategies, and pedagogical approaches that inform how you'll use this Embark Guide.

Identify your audience persona(s) - Determine whether your students align with:

-  **Code-free persona:** Students focusing on AI concepts, strategy, and no-code tools
-  **Code-based persona:** Students with programming backgrounds who will implement AI systems

Navigating the Content

Look for persona icons - Throughout the guide, you'll see both symbols indicating which content is most relevant for each audience. Content with both icons is applicable to both personas.

Use the modularized structure - The course is broken into discrete sections that can be:

- Delivered in sequence for a full 4-hour session
- Distributed across multiple class periods
- Selectively included based on your available time and student needs

Select appropriate hands-on activities:

- **Code-free faculty:** Direct students to Amazon Quick for no-code AI solution building
- **Engineering-focused faculty:** Use Amazon Kiro coding assistant IDE with Bedrock Agentic features
- **Science-focused faculty:** Leverage Amazon SageMaker Studio for model experimentation with Bedrock Agentic features

Adapting to Your Context

Time constraints: If you have less than 4 hours, prioritize the foundational concepts and select one hands-on activity that best matches your students' backgrounds.

Mixed audiences: For classes with both technical and non-technical students, consider parallel activities where different personas work with different tools on the same conceptual challenge.

Assessment flexibility: The guide supports various assessment approaches—adapt the suggested activities to match your institution's requirements and grading structure.

Additional Resources

Reference the course materials and supplementary documentation as needed. This guide is meant to complement, not replace, the detailed lesson plans and slide decks provided in the full course package.

Remember: The same fundamental Agentic AI concepts apply across all personas—what change is the depth of technical implementation and the tools used for hands-on practice.

Content Modularization

Lecture activities

All MLU EEP course content is modularized, allowing faculty members the flexibility to choose modules that are most relevant to their educational backgrounds.

Throughout our course's slide deck, we use two different icons to identify the proposed target persona relate to the content:



code-based Course Icon



code-free Course Icon

Both lecture content, supported by slide decks, and lab hands-on material are identified by these icons.

Usage example

If you your background is code-free, look for all slides that have this icon: 

otherwise, look for all slides that have this icon: 

Hands-on activities

Hands-on activities also are modularized considering three different personas, adding two categories to the technical personas:

code-free faculty

These hands-on activities aim to give experience creating AI solutions using Amazon Quick. This no-code environment is ideal for faculties without programming background, enabling them to build sophisticated AI solutions effortlessly.

Engineering focused faculty with software engineering path

These hands-on activities are suited for personas following a software engineering path but also can be used for general code-based faculty personas. The activities use Amazon Kiro coding assistant IDE to practice Agentic AI concepts by implementing Bedrock Agentic features. This environment is ideal for faculties with some coding experience, allowing them to explore how AI-powered coding assistants can accelerate development and help build robust AI systems efficiently.

Science focused faculty

These hands-on activities are suited for personas following a science path but also can be used for general code-based faculty personas. The activities use the Amazon SageMaker Studio environment to practice Agentic AI concepts by implementing Bedrock Agentic features. This environment provides an ideal workspace for data scientists experimenting on different AI models and architectures to address business challenges, without software and hardware infrastructure distractions.

Fitting the content to your class time frame

Based on the split recommendation outlined above, the figure below illustrates the proposed distribution of slides and labs between two audience types: code-based and code-free personas. The technical track contains all code-free content, as it encompasses all code-free material plus additional code-based depth.

The images below suggest a distribution of content for both personas (number of slides and lab notebooks)

Note: particularly for this module, as it presents Agentic AI fundamental concepts, the same content fits both code-based and code-free personas.

Depending on the number of slides and labs selected, you can choose to fit them into one or more classes.

Note: This is just an estimate considering a 2-hour class window. You can adapt it to your regular class time window, according to your activities and class dynamics.

Below we suggest how to separate the content for both personas:

- For code-free personas, the content fits a 2-hour window class
- For code-based personas, the content can split into two 2-hour window classes

